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STUDY OF NETWORK CABLES

a) Understand different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic cable:

Aim:

To understand the different type of cables used in networking and also make our own Ethernet cross-over cable / straight cable.

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/ use	Image.
UTP	category 3	10 bps	Ad: • cheaper in cost • Easy to install as they have a smaller overall diameter Disad: • More prone to (EMI) Electro magnetic interference and noise	10 Base-T ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
	category 5	Up to 100 Mbps			
	category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	Ad: • Shielded • Faster than UTP. • Less susceptible to noise and interference Disad: • Expensive • Greater installation and effort	Gigabit ethernet 10G Ethernet (55m) Widely used in data centres.	
SSTP	category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100 m)	

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Adv: • High band width • Immune to interference • Low loss bandwidth • versatile Disad • Limited distance • cost	speed of signal is 500m Television network High speed internet connections.	
fibre optics cable	single mode Multi mode	100 Gbps	Adv: • High speed • High bandwidth • High security • Long distance Dis: • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100 meters.	

b) Make Your Own Ethernet Cross-over cable/ Straight cable

- Ethernet cabling. CAT5e is certified for gigabit support but CAT5 cabling works as well; just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs
- Optional two plug shields.

Result:

Hence the different types of cables used in networking are understood.